

Le Gao

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EDUCATION

Harbin Institute of Technology
Integrated Circuit Science and Engineering

Ph.D. (Direct M.S.–Ph.D.)
Sep. 2022 – Jun. 2027

Harbin University of Science and Technology

Bachelor
Sep. 2018 – June 2022

Honors

- Special Scholarship
- Outstanding Student Leader (×2)
- China National Scholarship (Doctoral)
- Second-Class Scholarship
- Outstanding Youth League Member
- Heilongjiang Provincial Merit Student

RESEARCH INTERESTS

Physics-informed machine learning (PIML) for remaining useful life (RUL) prediction of power electronic devices.

TECHNICAL SKILLS

Programming: Python, LabVIEW, MATLAB, Saber, LaTeX

Experimental Skills: Space-environment irradiation testing; designed and built a power-cycling test bench; developed a real-time device-health monitoring system.

ACADEMIC SERVICE

- Reviewer for *IEEE Transactions on Power Electronics* (IEEE TPEL).
- Reviewer for *IEEE Journal of Emerging and Selected Topics in Power Electronics* (IEEE JESTPE).

JOURNAL ARTICLES

- **L. Gao**, C. Liu, F. Wu, *et al.*, “Prediction method for the remaining useful life of power MOSFETs based on deep learning,” *IEEE Trans. Power Electron.*, doi: [10.1109/TPEL.2025.3586684](https://doi.org/10.1109/TPEL.2025.3586684).
- **L. Gao**, C. Liu, Y. Xiao, *et al.*, “Remaining useful life prediction of power electronic devices with physics-informed deep learning and sparse data,” *IEEE Trans. Power Electron.*, doi: [10.1109/TPEL.2025.3563853](https://doi.org/10.1109/TPEL.2025.3563853).
- **L. Gao**, M. Ren, C. Liu, “A review of data-driven reliability prediction for power MOSFETs,” *Microelectronics* (in Chinese), vol. 54, no. 4, pp. 644–652, 2025.
- **L. Gao**, C. Liu, M. Huo, *et al.*, “Early-stage remaining useful life prediction for power electronic devices via deep learning with physical guidance,” *submitted to IEEE Trans. Power Electron.* (under review).
- **L. Gao**, C. Liu, M. Huo, *et al.*, “Uncertainty-aware physics-informed deep learning for remaining useful life prediction of power electronic devices,” (manuscript in preparation).

CONFERENCE PRESENTATIONS

- **L. Gao** (2026). “*Uncertainty-Aware Physics-Informed Machine Learning for Remaining Useful Life Prediction of Power Electronic Devices*.” Oral presentation at the 2026 *Applied Power Electronics Conference and Exposition* (APEC) Conference , **America**.
- **L. Gao** (2025). “*Remaining Useful Life Prediction of Power Electronic Devices via Physics-Informed Deep Learning in Radiation Environments*.” Oral presentation at the 2025 *Radiation and its Effects on Components and Systems* (RADECS) Conference , **Belgium** (the most influential conference on radiation effects and reliability).
- **L. Gao** (2025). “*Remaining Useful Life Prediction of Power Electronic Devices with Physics-Informed Deep Learning under Sparse-Data Conditions*.” Oral presentation at the 6th *International Conference on Radiation Effects of Electronic Devices*.
- **L. Gao** (2024). “*Machine-Learning-Based Radiation Reliability Degradation Modeling of SiC MOSFETs*.” Oral presentation at the 6th *China Radiation Physics Academic Exchange Conference*.

PATENTS AND SOFTWARE COPYRIGHTS

- **L. Gao** (2/8), ZL202410400889.6 “Reliability Prediction System for Electronic Devices,” granted.
- **L. Gao** (2/7), ZL202410177294.9 “Reliability Prediction Method for Power MOSFETs,” granted.
- **L. Gao** (2/7), CN202311312876.5 “Reliability Prediction Method for Schottky Barrier Diodes,” pending.
- **L. Gao** (2/7), CN202311317217.0 “Modeling Method for Power MOSFET Reliability,” pending.
- **L. Gao** (2/8), CN202410829095.1 “Power-Cycling Test System and Method for Electronic Components,” pending.
- **L. Gao** (2/7), CN202510499482.8 “Multi-Scale Life Prediction of Electronic Components via Physics-Based Deep Learning,” pending.
- **L. Gao** (2/6), CN202510514888.9 “Life Prediction and Health Management System/Method/Device/Medium for Key Components in Switching Power Supplies,” pending.
- **L. Gao** (2/7), Application No. 2025109542691 “Early Remaining Useful Life Prediction for Power Electronic Devices,” pending.
- **L. Gao** (2/7), Application No. 2025116145441 “Lifetime Prediction Method for Power Electronic Devices with Uncertainty Quantification,” pending.
- **L. Gao** (3/7), CN202311312872.7 “Reliability Prediction Method for IGBT Devices,” pending.
- **L. Gao** (4/7), CN202510555263.7 “Reliability Evaluation of Deep Neural Networks via Analytical and Hybrid Fault Injection—Method/System/Device/Medium,” pending.
- **L. Gao** (5/6), CN202310770210.8 “Reliability Analysis Method for Transistors Based on FEDformer,” pending.
- **L. Gao** (6/6), CN202310773439.7 “Reliability Analysis Method for Diodes Based on Pyraformer,” pending.
- **L. Gao**, CR No. 2024SR0241261 “Reliability-Prediction Software for Electronic Components,” registered 2024.
- **L. Gao**, CR No. 2024SR0684147 “Power-Cycling Test System for Electronic Components,” registered 2024.

PROJECTS

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| 1. Reliability Assurance for Astronautic Electronic Information Systems
<i>(Defense Basic Research)Conducted experiments using ground-based space-environment simulators</i> | Key contributor |
| 2. Reliability Prognostics and Optimization for Discrete Semiconductor Devices
<i>(Equipment Pre-Research)Performed reliability-parameter optimization and lifetime prediction</i> | Key contributor |
| 3. Failure Mechanisms and Lifetime Prediction Methods for Ultra-Wide-Bandgap Devices
<i>(NSFC)Analyzed device failure mechanisms; carried out early-life lifetime modeling and uncertainty analysis</i> | Key contributor |
| 4. Modeling and Accuracy Verification of Operational-Amplifier Electrical Characteristics
<i>(Equipment Pre-Research)Built simulation models in Saber</i> | Key contributor |

EXPERIENCE

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| Chair , Graduate Student Union
<i>Harbin Institute of Technology</i> | Sep. 2022 – Jun. 2023 |
| Secretary, Integrated Circuit Graduate Party Branch
<i>Harbin Institute of Technology</i> | Oct. 2025 – Present |
| Delegate, 15 th –16 th HIT Graduate Student Congress
<i>Harbin Institute of Technology</i> | Sep. 2022 – Jun. 2023 |